Boundary Controllability of Linearized Compressible Navier-Stokes Equations in 1D using One Control

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Abstract

This poster aims to summarize the boundary controllability results for the linearized compressible Navier-Stokes system in one dimension. This linearized system consists of a transport (density) equation coupled with a heat (velocity) equation (with first-order coupling). We consider cases where only one boundary control can act in the density or velocity component at one end of the domain or the difference between the values at both ends. In this setup, we state the existing results in the literature and some new results which we have obtained. We see that these controllability results are optimal concerning the regularity of initial states (in the velocity case) and time (in the density case). Finally, this poster ends with some future directions of research.

The Compressible Navier-Stokes System

The Navier–Stokes equations for a viscous compressible isentropic fluid in an interval consists of the equation of continuity and the momentum equation

$$\rho_t + (\rho u)_x = 0,$$
$$\rho[u_t + uu_x] + p_x(t,x) - \nu u_{xx} = 0.$$

Here

- $\rho = \rho(t,x)$ denotes the density of the fluid,
- $u = u(t,x)$ is the velocity,
- λ, μ are called the viscosity coefficients that satisfy $\mu > 0, \lambda + \mu \geq 0$.
- The pressure p satisfies the following constitutive equation

$$p = a\rho^\gamma, \quad a > 0, \gamma \geq 1.$$

The Linearized System

- Linearized system around (Q_0, V_0) with $Q_0, V_0 > 0$ is

$$\text{(1)} \qquad \begin{cases} \rho_t + V_0\rho_x + Q_0 u_x = 0 & \text{in } (0,T)\times(0,L), \\ u_t - \nu_0 u_{xx} + V_0 u_x + b\rho_x = 0 & \text{in } (0,T)\times(0,L). \end{cases}$$

- Here $T, L > 0$, $\nu_0 = \frac{\nu}{Q_0}$ and $b = a\gamma Q_0^{\gamma-2}$.
- The Initial state is given by

$$\text{(2)} \qquad \rho(0,x) = \rho_0(x), \quad u(0,x) = u_0(x), \quad x \in (0,L).$$

- We consider one of the following boundary conditions:

$$\begin{cases} \rho(t,0) = \rho(t,L),\ u(t,0) = u(t,L),\ u_x(t,0) = u_x(t,L), \text{ for } t \in (0,T). \\ \rho(t,0) = 0, \quad u(t,0) = 0, \quad u(t,L) = 0, \quad \text{for } t \in (0,T). \\ \rho(t,0) = \rho(t,L), \quad u(t,0) = 0, \quad u(t,L) = 0, \text{ for } t \in (0,T). \end{cases}$$

Existing Results

- Consider the following control problem:

$$\text{(3)} \begin{cases} \rho_t + V_0\rho_x + Q_0 u_x = 0, & \text{in } (0,T)\times(0,2\pi), \\ u_t - \nu_0 u_{xx} + V_0 u_x + b\rho_x = 0, & \text{in } (0,T)\times(0,2\pi), \\ \rho(t,0) = \rho(t,2\pi), & \text{for } t \in (0,T), \\ u(t,0) = u(t,2\pi) + q(t),\ u_x(t,0) = u_x(t,2\pi), & \text{for } t \in (0,T), \\ \rho(0,x) = \rho_0(x), \quad u(0,x) = u_0(x), & x \in (0,2\pi). \end{cases}$$

- Here $q \in L^2(0,T)$ is a boundary control.

Definition 1 (Null Controllability)
Let H be a Hilbert space. We say the system (3) is null controllable at time T in H if for given $(\rho_0, u_0) \in H$, there exists a $q \in L^2(0,T)$ such that the solution (ρ, u) of (3) satisfies

$$(\rho(T), u(T)) = (0,0).$$

Theorem 1 (S. Chowdhury, D. Mitra; JEE 2015) *Let $s > \frac{9}{2}$ be given. Then the system (3) is null controllable at time $T > \frac{2\pi}{V_0}$ in $\dot{H}^{s+1}_{per}(0,2\pi) \times \dot{H}^s_{per}(0,2\pi)$.*

Here the dot represents that the functions has mean value zero.

Theorem 2 (D. Maity; ESAIM: COCV 2015) *Let $0 < T < \frac{2\pi}{V_0}$ be given. Then the system (3) is not null controllable at time T in $(L^2(0,2\pi))^2$.*

New Results (Control on Density)

$$\text{(4)} \begin{cases} \rho_t + V_0\rho_x + Q_0 u_x = 0, \\ u_t - \nu_0 u_{xx} + V_0 u_x + b\rho_x = 0, \\ \rho(t,0) = \rho(t,2\pi) + p_1(t), \\ u(t,0) = u(t,2\pi),\ u_x(t,0) = u_x(t,2\pi), \\ \rho(0,x) = \rho_0(x), \quad u(0,x) = u_0(x), \end{cases}$$

for $(t,x) \in (0,T)\times(0,2\pi)$ and $p_1 \in L^2(0,T)$ is the boundary control.

$$\text{(5)} \begin{cases} \rho_t + \rho_x + cu_x = 0, \\ u_t - u_{xx} + u_x + c\rho_x = 0, \\ \rho(t,0) = p_2(t), \\ u(t,0) = 0,\ u(t,1) = 0, \\ \rho(0,x) = \rho_0(x), \quad u(0,x) = u_0(x), \end{cases}$$

for $(t,x) \in (0,T)\times(0,1)$ and $p_2 \in L^2(0,T)$ is the boundary control.

Theorem 3 (—, 2024)
(i) The system (4) is null controllable at any time $T > \frac{2\pi}{V_0}$ in $(\dot{L}^2(0,2\pi))^2$ if and only if

$$\frac{2\sqrt{bQ_0 - V_0^2}}{\nu_0} \notin \mathbb{N}.$$

(ii) If $0 < T < \frac{2\pi}{V_0}$, the system (4) is not null controllable at time T in $(\dot{L}^2(0,2\pi))^2$.

Theorem 4 (K. Bhandari, S. Chowdhury, R. Dutta, —, 2024)
(i) Let $c \in (0,4]$ and $T > 1$. Then the system (5) is null controllable at time T in $\dot{L}^2(0,1) \times L^2(0,1)$.
(ii) If $0 < T < 1$, the system (5) is not null controllable at time T in $(L^2(0,1))^2$.

New Results (Control on Velocity)

$$\text{(6)} \begin{cases} \rho_t + V_0\rho_x + Q_0 u_x = 0, \\ u_t - \nu_0 u_{xx} + V_0 u_x + b\rho_x = 0, \\ \rho(t,0) = \rho(t,2\pi),\ u(t,0) = u(t,2\pi) + q_1(t), \\ u_x(t,0) = u_x(t,2\pi), \\ \rho(0,x) = \rho_0(x), \quad u(0,x) = u_0(x), \end{cases}$$

for $(t,x) \in (0,T)\times(0,2\pi)$ and $q_1 \in L^2(0,T)$ is the boundary control.

$$\text{(7)} \begin{cases} \rho_t + \rho_x + cu_x = 0, \\ u_t - u_{xx} + u_x + c\rho_x = 0, \\ \rho(t,0) = \rho(t,1), \\ u(t,0) = 0,\ u(t,1) = q_2(t), \\ \rho(0,x) = \rho_0(x), \quad u(0,x) = u_0(x), \end{cases}$$

for $(t,x) \in (0,T)\times(0,1)$ and $q_2 \in L^2(0,T)$ is the boundary control.

Theorem 5 (—, 2024)
(i) The system (6) is null controllable at any time $T > \frac{2\pi}{V_0}$ in $\dot{H}^1(0,2\pi) \times \dot{L}^2(0,2\pi)$ if and only if

$$\frac{2\sqrt{bQ_0 - V_0^2}}{\nu_0} \notin \mathbb{N}.$$

(ii) If $0 \leq s < 1$, the system (6) is not null controllable at any time $T > 0$ in $\dot{H}^s_{per}(0,2\pi) \times \dot{L}^2(0,2\pi)$.

Theorem 6 (K. Bhandari, S. Chowdhury, R. Dutta, —, 2024)
(i) Let $T > 1$ be given. Then there exists a countable set $\mathcal{N}$ such that for $c \in (0,4] \setminus \mathcal{N}$, the system (7) is null controllable at time T in the space $\dot{H}^{\frac{1}{2}}(0,1) \times L^2(0,1)$.
(ii) If $0 \leq s < \frac{1}{2}$, the system (7) is not null controllable at any time $T > 0$ in $\dot{H}^s(0,1) \times L^2(0,1)$.

Key Points

- **Tools:** Spectral analysis, method of moments and Ingham-type inequality.
- If $\frac{2\sqrt{bQ_0 - V_0^2}}{\nu_0} \in \mathbb{N}$, then the associated linearized operator of the system (4) (or (6)) admits an eigenvalue with algebraic multiplicity= geometric multiplicity= 2.
- The restriction on c and the set $\mathcal{N}$ appears while proving the observation terms are non-zero.
- Null controllability of (1)-(2) at large time T using a Dirichlet control on velocity $q_3 \in L^2(0,T)$, given as $\rho(t,0) = 0, \quad u(t,0) = 0, \quad u(t,1) = q_3(t), \quad t \in (0,T)$, is still an open problem.

References

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